



COMPETENCY STANDARD

FOR

**ADVANCED CNC MACHINING CENTER
OPERATION AND PRACTICE**

(Light Engineering Sector)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Advanced CNC Machining Center Operation and Practice serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the Light Engineering sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for Advance CNC Machining Center Operation and Practice. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.

- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-ACNC-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS Policies and Procedures 2. Practice Personal Health and Safety Procedures 3. Report Hazards and Risks 4. Respond to Emergencies	10
SICIP-LE-ACNC-02-G	Operate in a Self-Directed Team	1. Identify Team Goals and Work Processes 2. Communicate and Cooperate with Team Members. 3. Work as a Team Member. 4. Solve Problems as a Team Member	10
Total Hour			20

Sector Specific Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-ACNC-01-S	Apply Green Practices in Light Engineering Sector	1. Interpret Green Concepts 2. Minimize Resource Use in the Workplace 3. Implement Waste Management Practices	10
SICIP-LE-ACNC-02-S	Understand Mechanical Drawing	1. Interpret Mechanical Drawing 2. Create Mechanical Drawing	10
SICIP-LE-ACNC-03-S	Work with Hand Tools and Power Tools	1. Inspect Hand Tools and Power Tools for Usability 2. Use Hand Tools Properly and Safely 3. Operate Power Tools Properly and Safely 4. Clean/Maintain Hand Tools and Power Tools After Use	10
Total Hour			30

Occupation Specific Competencies (250 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-ACNC-01-O	Operate CNC Machining Center	1. Prepare CNC Machining Center 2. Select Cutting Tools 3. Perform CNC Machining Center using Controller	30
SICIP-LE-ACNC-02-O	Operate CNC Turning Center	1. Prepare CNC Turning Center 2. Select Cutting Tools 3. Perform CNC Turning Center using Controller	30
SICIP-LE-ACNC-03-O	Create Program using G & M Code for CNC Machine	1. Determine Machine Co-Ordinate Systems 2. Program CNC Machining Center using G & M Code 3. Program CNC Turning Center using G & M Code 4. Test and Debug CNC Program	40
SICIP-LE-ACNC-04-O	Perform 2D Machining Operation using CNC Machining Center	1. Set-up Machine and Workpiece 2. Create/input Program and Simulate the Program 3. Operate CNC Machining Center for 2D Machining 4. Perform Basic CNC Maintenance	65
SICIP-LE-ACNC-05-O	Perform 2D Machining Operation using CNC Turning Center	1. Set-up Machine Cutting tools and Workpiece 2. Create/input Program and Simulate the Program 3. Operate CNC Turning Center for 2D Machining	65
SICIP-LE-ACNC-06-O	Interpret Co-Ordinate Measuring Machine (CMM)	1. Understand the Concept of CMM 2. Understand CMM Program and Operation 3. Analyze Data	20
Total Hour			0 Hrs.

COMPETENCY STANDARD: ADVANCED CNC MACHINING CENTER OPERATION AND PRACTICE

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-ACNC-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
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	May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations 1.3 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards 3.2 Biological hazards 3.3 Physical Hazards 3.3.1 Machine hazards 3.3.2 Materials hazards 3.3.3 Tools and Equipment hazards
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures 1.2 Work safety procedures 1.3 Emergency procedures 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Explosion response 1.3.4 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 Types of PPE and their uses 1.6 Personal hygiene practices 1.7 OHS awareness
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Applying personal work safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergency procedures

	2.5 Maintaining physical well-being in the workplace 2.6 Using firefighting accessories and fire extinguishers 2.7 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning/operational manual 4.3 Monitoring tools/equipment (CCTV, sensor, alarm, etc.) 4.4 PPEs 4.5 Firefighting equipment 4.6 Emergency response manual 4.7 Emergency Systems (Fire alarms, emergency exits, etc.) 4.8 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 applied OHS policies and procedures 1.2 selected and used the PPE 1.3 practiced personal health and safety procedures 1.4 reported incidents arising from hazards and risks to the authority 1.5 implemented plans and procedures to respond emergency 1.6 applied basic first aid procedure
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A SELF-DIRECTED TEAM	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-ACNC-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members.	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practices. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range May Include but not limited to:
1. Forms of communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective formal and informal forms of communication 1.5 Value of diversity in team functioning 1.6 Correct use of workplace terminology 1.7 Team's duties, responsibilities, authorities, objectives and task requirements 1.8 Support mechanism to other members of team to ensure achievements of goals 1.9 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.4 Using formal and informal forms of communication 2.5 Understanding and valuing views and opinions of other team members 2.6 Performing tasks in accordance with organizational and team requirements and workplace procedures & practices 2.7 Identifying current and potential problems faced by a team 2.8 Identifying solutions to the problem 2.9 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities

	3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Work books 4.3 Learning manuals 4.4 Flexible organizational policies 4.5 Communication platforms 4.6 Performance tracking tools

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 described roles and responsibilities of team members 1.3 communicated and cooperated with team members 1.4 explained formal and informal forms of communication 1.5 worked as an active team member 1.6 solved problems effectively as a team member
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

Sector Specific Competencies

Unit of Competency: APPLY GREEN PRACTICES IN LIGHT ENGINEERING SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-ACNC-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in light engineering sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Basic concepts of green</u> are understood. 1.2 Energy efficiency principles are interpreted. 1.3 <u>Sustainable materials</u> are identified. 1.4 Pollution control methods are understood. 1.5 <u>Environmental certifications and standards</u> are identified. 1.6 <u>Green supply chains</u> are interpreted to minimize environmental impact.
2. Minimize resources use in the workplace	2.1 Resource consumption is identified in the light engineering sector. 2.2 <u>Energy-efficient equipment</u> is identified and maintained. 2.3 Materials are sourced ensuring that sustainable and recyclable materials are prioritized. 2.4 Water usage is minimized through efficient management. 2.5 Production processes are optimized. 2.6 <u>Green technologies and innovations</u> are implemented in the workplace.
3. Implement waste management practices	3.1 <u>Waste types</u> identified in the light engineering sector. 3.2 Waste segregation practices are interpreted. 3.3 Recycling programs are established through external recycling facilities. 3.4 <u>Waste reduction strategies</u> are understood. 3.5 Safe disposal practices are described. 3.6 <u>Waste-to-energy solutions</u> are explored to reduce landfill contributions.

Range of Variables

Variable	Range
	May include but not limited to:
1. Basic concepts of green	1.1 Principles of Green Practices 1.2 Sources of Environmental Impacts 1.3 Resource Utilization 1.4 Sustainable Alternatives 1.5 Waste Management Practices 1.6 Green Habits at Work 1.7 Compliance and Regulations
2. Sustainable materials	2.1 Recycled Metals 2.2 Biodegradable Lubricants and Fluids 2.3 Composite Materials with Low Environmental Impact 2.4 Low-VOC Paints and Coatings 2.5 Eco-friendly Packaging Materials 2.6 Durable and Repairable Components
3. Environmental certifications and standards	3.1 ISO 14001: Environmental Management System (EMS) 3.2 LEED Certification (Leadership in Energy and Environmental Design) 3.3 RoHS (Restriction of Hazardous Substances) 3.4 REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) 3.5 Energy Star Certification 3.6 Eco-Labeling Standards 3.7 Green Building Codes & Local Environmental Compliance
4. Green supply chains	4.1 Environmentally Responsible Sourcing 4.2 Eco-Friendly Transportation and Logistics 4.3 Supplier Environmental Compliance 4.4 Sustainable Packaging and Distribution
5. Energy-efficient equipment	5.1 High-Efficiency Electric Motors (IE2/IE3/IE4 Rated) 5.2 Variable Frequency Drives (VFDs) 5.3 Energy-Efficient LED Lighting Systems 5.4 Energy-Star Rated Industrial Equipment 5.5 Heat Recovery and Insulation Systems
6. Green technologies and innovations	6.1 Renewable Energy Integration 6.2 Additive Manufacturing (3D Printing) 6.3 Smart Manufacturing Systems (Industry 4.0) 6.4 Waste-to-Energy Technologies 6.5 Low-Emission Welding and Cutting Equipment 6.6 Closed-Loop Cooling and Lubrication Systems 6.7 Eco-Innovation in Product Design
7. Waste types	7.1 Metal Waste 7.2 Plastic and Packaging Waste 7.3 Hazardous Waste

	7.4 Electronic Waste (E-Waste) 7.5 Dust and Particulate Waste 7.6 Wood and Pallet Waste
8. Waste reduction strategies	8.1 5S Workplace Organization 8.2 Lean Manufacturing Practices 8.3 Material Optimization and Nesting Techniques 8.4 Reusing and Repurposing Waste
9. Waste-to-energy solutions	9.1 Thermal Conversion (Incineration with Energy Recovery) 9.2 Gasification and Pyrolysis 9.3 Biomass Boilers Using Organic Waste 9.4 Use of Waste Oils as Alternative Fuel 9.5 Waste Heat Recovery Systems (WHRS) 9.6 On-Site Micro Energy Generation Units

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of pollution. 1.2 Pollution control technologies 1.3 Legal and environmental frameworks: overview of national environmental policies and local environmental compliance regulations. 1.4 Importance of ISO 14001 and other EMS (Environmental Management System) standards. 1.5 Classification of sustainable, recyclable, and hazardous materials used in production. 1.6 Strategies for minimizing water and energy use during material processing. 1.7 Waste hierarchy: reduce, reuse, recycle, recover, and disposal (5Rs). 1.8 Industrial waste-to-energy concepts.
2. Underpinning Skills	2.1 Measuring and analyzing energy and water consumption using meters. 2.2 Identifying and operating energy-efficient machinery. 2.3 Monitoring and logging resource consumption data. 2.4 Adjusting machine settings or workflow layouts. 2.5 Applying green inventory practices. 2.6 Segregating waste effectively into recyclable, hazardous, and general categories. 2.7 Implementing basic pollution control activities.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the

	workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: UNDERSTAND MECHANICAL DRAWING	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-ACNC-02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand mechanical drawing. It specifically includes the tasks of interpreting mechanical drawing and creating mechanical drawing.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret mechanical drawing	1.1 Different types of <u>mechanical drawings</u> are interpreted. 1.2 Different <u>views of mechanical drawing</u> are identified and described. 1.3 Necessity of putting dimensions in different drawings is comprehended.
2. Create mechanical drawing	2.1 Drawing is created accurately. 2.2 Components, objects, parts, materials and assemblies are identified, drawn and described. 2.3 Base line or datum points are specified, as required. 2.4 Instructions are included in drawing. 2.5 <u>Tolerance</u> , limits and fits are identified in technical drawing.

Range of Variables

Variable	Range
	May include but not limited to:
1. Mechanical drawings	1.1 Sketch components 1.1.1 Line 1.1.2 Rectangle 1.1.3 Circle 1.1.4 Ellipse 1.1.5 Polygon 1.1.6 Arc 1.2 Elements to determine dimension 1.2.1 Central line 1.2.2 Object line 1.2.3 Tolerance 1.2.4 Ladder line 1.2.5 Extension line 1.2.6 Hidden line 1.2.7 Cutting plane line 1.2.8 Short break

	1.2.9 Long brake
2. Views of mechanical drawing	2.1 Top and Bottom View 2.2 Right and Left View 2.3 Front View 2.4 Sectional view 2.5 Isometric View 2.6 Oblique View 2.7 Hidden view 2.8 Perspective view
3. Tolerance	3.1 General 3.2 Angular 3.3 Geometric

Curricular Content Guide

1. Underpinning knowledge	1.1 Drawing tools and supplies 1.2 Drawing symbols 1.3 Different views of drawing 1.4 Dimensioning techniques 1.5 Tolerance, limits and fits 1.6 Engineering materials
2. Underpinning skills	2.1 Interpreting mechanical drawing 2.2 Creating mechanical drawing
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Drawing tools and supplies 4.4 Specimen parts and components 4.5 Drawings and sketches

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted mechanical drawing 1.2 created mechanical drawing
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2. Methods of assessment	Methods of assessment may include but is not limited to: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK WITH HAND TOOLS AND POWER TOOLS	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-ACNC-03-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work with hand tools and power tools properly. It specifically includes the tasks of inspecting hand tools and power tools for usability, using hand tools properly and safely, operating power tools properly and safely, and cleaning/maintaining hand tools and power tools after use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect hand tools and power tools for usability	1.1 Appropriate tools are selected. 1.2 Application of tools to job requirement is determined. 1.3 Usability of tools are checked and verified. 1.4 <u>Hand tools</u> and <u>power tools</u> are prepared. 1.5 Sources of power supply for power tools identified.
2. Use hand tools properly and safely	2.1 Appropriate hand tool for the job is used. 2.2 Proper and safe use/operation is applied in the different types of hand tools. 2.3 <u>Safety precautions</u> are observed when using hand tools. 2.4 Unsafe or faulty tools are identified and marked for repairing.
3. Operate power tools properly and safely	3.1 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.2 Proper sequence of operation is applied in using power tools to produce results. 3.3 Power tools are used safely in accordance with manufacturer's operating specification.
4. Clean/maintain hand tools and power tools after use	4.1 Dust and foreign materials are removed from power tools in accordance with workplace standard. 4.2 Scraps or wastages are removed and cleaned 4.3 Condition of tools is checked after use. 4.4 Appropriate lubricant is applied after use and prior to storage. 4.5 <u>Measuring tools</u> are checked and calibrated. 4.6 Defective tools, instruments, power tools and accessories are inspected and corrected or replaced.

Range of Variables

Variable	Range May include but not limited to:
1. Hand tools	1.1 File 1.2 Adjustable Wrench 1.3 Pneumatic Wrenches 1.4 Leveler 1.5 Allen Key 1.6 Hacksaw (for cutting conduit) 1.7 Ball peen hammer 1.8 Socket Wrench 1.9 Ratchet 1.10 Soft hammer
2. Power tools	2.1 Hand grinding machine 2.2 Power Saw 2.3 Power Drills 2.4 Angle Grinders and Grinders 2.5 Pedestal Drills
3. Safety precautions	3.1 Use of appropriate PPEs 3.2 Proper hand, feet and eye coordination 3.3 Working environment 3.4 Safe operating condition of hand tools and power tools 3.5 Awareness to OHS requirements
4. Measuring tools	4.1 Measuring tape 4.2 Vernier Calipers 4.3 Calipers 4.4 Micrometer 4.5 Thread Gauge 4.6 Filler Gauge 4.7 Dial Indicator 4.8 Bevel Protractor 4.9 Bore Gauge 4.10 Radius gauge

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of tools, functions and use 1.2 Types of Hand tools and their proper use and techniques 1.3 Types of Power tools, use and safe handling method 1.4 Technical application of tools 1.5 Procedures in the use of hand tools and power tools 1.6 Policies and procedures for occupational health and safety 1.7 Use of PPE 1.8 Handling of tools and equipment
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	1.9 Reporting and documentation 1.10 Preventive maintenance 1.11 Methods and techniques 1.12 Quality procedures 1.13 Waste management 1.14 Storage procedures
2. Underpinning Skills	2.1 Using appropriate hand tool for the job 2.2 Observing safety precautions when using hand tools 2.3 Using power tools correctly and safely in accordance to manufacturer's operating specification. 2.4 Checking condition of tools after use 2.5 Applying appropriate lubricant on hand tools and power tools after use and prior to storage 2.6 Inspecting and correcting or replacing defective tools, instruments, power tools and accessories 2.7 Storing Tools and power tools safely in appropriate location
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Light Engineering hand tools and power tools 4.3 Pens 4.4 Papers 4.5 Work books 4.6 Operation and maintenance manuals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Used appropriate hand tools for the job. 1.2 Observed safety precautions when using hand tools. 1.3 Used power tools safely in accordance to manufacturer's operating specification. 1.4 Cleaned and maintained hand tools and power tools after use and prior to storage. 1.5 Inspected and corrected or replaced defective tools, instruments, power tools and accessories. 1.6 Stored tools and power tools safely in appropriate
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	location.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation-specific Competencies

Unit of Competency: OPERATE CNC MACHINING CENTER	Nominal Duration: 30 hrs.	Unit Code: SICIP-LE-ACNC-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate CNC machining center. It specifically includes the tasks of preparing CNC machining center, selecting cutting tools and performing CNC machining center using controller.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare CNC Machining Center	1.1 Safety protocols and <u>PPE</u> requirements are reviewed and followed before starting any machine operation. 1.2 <u>Types of CNC machining center</u> and an overview of their <u>components</u> are understood and identified. 1.3 Air pressure, coolant level, and lubrication oil are checked to ensure proper machine functioning. 1.4 CNC machining center is powered on and off according to standard operating procedures. 1.5 ATC (Automatic Tool Change) function is performed to ensure proper tool handling during machining. 1.6 The workpiece is clamped securely using the appropriate clamping kits.
2. Select Cutting Tools	2.1 Various types of <u>cutting tools</u> and <u>inserts</u> are identified based on the material. 2.2 Appropriate sequence of cutting tools for the machining is selected. 2.3 Tool and tool holder are loaded onto the machine to ensure stability during the machining process. 2.4 Appropriate <u>cutting parameters</u> are followed according to the job material specifications.
3. Perform CNC Machining Center using Controller	3.1 CNC machine axes positioning is performed using the Manual Pulse Generator (MPG). 3.2 Control panel main buttons and soft keys are used for machine adjustments. 3.3 Machine panel buttons and switches are identified and selected. 3.4 MDI mode is executed and created as required for the operation. 3.5 <u>Memory mode</u> is executed. 3.6 <u>Overrides keys</u> are performed. 3.7 Work offset and tool offset are set and adjusted.

Range of Variables

Variable	Range May include but not limited to:
1. PPE	1.1 Apron 1.2 Mask 1.3 Safety goggles 1.4 Ear plug 1.5 Helmet 1.6 Safety shoes
2. Types of CNC machining center	2.1 3-Axis machine 2.2 5-Axis machine
3. Components	3.1 Controller 3.2 Spindle head 3.3 Tool magazine 3.4 Table 3.5 Base 3.6 Coolant system 3.7 Chip conveyor system
4. Cutting tools	4.1 Face mill 4.2 End mill 4.3 Ball mill 4.4 Bull mill 4.5 Chamfer mill 4.6 Machine tap 4.7 Drill bit
5. Inserts	5.1 Round 5.2 Square 5.3 Triangular
6. Cutting parameters	6.1 Cutting speed 6.2 Spindle speed 6.3 Feed rate 6.4 Depth of cut
7. Memory mode	7.1 Single block 7.2 Dry run 7.3 Program starts 7.4 Program stops
8. Overrides keys	8.1 Spindel speed overrides 8.2 Feed 8.3 Rapid

Curricular Content Guide

1. Underpinning knowledge	1.1 Concept of workplace safety protocols 1.2 Identify different types of CNC machining centers and their major components. 1.3 Understand the role of air pressure, coolant level, and
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	lubrication oil in maintaining machine performance. 1.4 Interpret standard operating procedures (SOPs) for powering CNC machines on and off safely. 1.5 Concept of Automatic Tool Changer (ATC) and its function in efficient tool handling. 1.6 Identify different cutting tools, tool inserts, and their applications for specific materials. 1.7 Understand cutting parameters (speed, feed, depth of cut) based on material specifications. 1.8 Identify and interpret CNC machine axes positioning methods using Manual Pulse Generator (MPG). 1.9 Concept of MDI (Manual Data Input) and Memory modes for CNC program execution.
2. Underpinning skills	2.1 Applying safety protocols and use PPE before operating machines. 2.2 Checking air pressure, coolant, and lubrication levels for proper machine function. 2.3 Performing machine power on and off following standard procedures. 2.4 Executing Automatic Tool Change (ATC) for proper tool handling. 2.5 Clamping workpieces securely using appropriate clamping kits. 2.6 Following appropriate cutting parameters according to job specifications. 2.7 Operating control panel buttons and soft keys for machine adjustments. 2.8 Executing MDI mode commands and create programs as required. 2.9 Executing memory mode for machining operations. 2.10 Performing overrides and adjustments using override keys.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Tools & equipment

	4.4 Job specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 prepared CNC machining center 1.2 selected cutting tools 1.3 performed CNC machining center using controller
2. Methods of assessment	Methods of assessment should include: 2.1 Written Test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE CNC TURNING CENTER	Nominal Duration: 30 hrs.	Unit Code: SICIP-LE-ACNC-02-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to operate CNC turning center. It specifically includes the tasks of preparing CNC turning center, selecting cutting tools and performing CNC turning center using controller.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare CNC Turning Center	1.1 <u>Major components of the CNC turning center</u> are identified and understood. 1.2 Air pressure, coolant level, lubrication, and power supply are checked. 1.3 CNC turning center is powered on and off following the standard procedures. 1.4 Hand jog and ATC (Automatic Tool Change) functions are performed. 1.5 Workpiece setup and tool setup are completed.
2. Select Cutting Tools	2.1 <u>Cutting tools</u> and <u>inserts</u> are identified and selected according to the machining requirements. 2.2 Appropriate sequence of cutting tools for the operation is selected. 2.3 Cutting tool geometry is explained to the operation. 2.4 Appropriate <u>cutting parameters</u> are followed according to the job material specifications.
3. Perform CNC Turning Center using Controller	3.1 <u>Functions keys</u> and <u>display keys</u> are used for machine adjustments. 3.2 <u>Mode key</u> and <u>override keys</u> are selected and used as per requirement. 3.3 CNC machine axes positioning is performed using the hand jog. 3.4 Work offset and tool offset are set and adjusted. 3.5 Visual programming system (VPS) and Shape creator are carried out as per drawing.

Range of Variables

Variable	Range
	May include but not limited to:
1. Major components	1.1 Controller

of the CNC turning center	1.2 Head stock 1.3 Tail stock 1.4 Tool turret 1.5 Coolant system 1.6 Chip conveyor
2. Cutting tools	2.1 Facing tool 2.2 Turning tool 2.3 Grooving/ parting tool 2.4 Threading tool 2.5 Boring tool 2.6 Drill bit 2.7 Center drill
3. Inserts	3.1 CNMG 3.2 DNMG 3.3 TNMG 3.4 VNMG 3.5 WNMG 3.6 Threading insert 3.7 Parting insert
4. Cutting parameters	4.1 Cutting speed 4.2 Spindle speed 4.3 Feed Rate 4.4 Depth of cut
5. Functions keys	5.1 Reset 5.2 Power up 5.3 Recover 5.4 F1-F4 5.5 X-dia measure 5.6 Next tool 5.7 Tool release 5.8 Z-face measure
6. Display keys	6.1 Program 6.2 Position 6.3 Offset 6.4 Current command 6.5 Alarms 6.6 Diagnostic 6.7 Setting 6.8 Help
7. Mode key	7.1 Edit 7.2 Memory 7.3 MDI 7.4 Handle job 7.5 Zero return 7.6 List program
8. Override keys	8.1 Feed rate 8.2 Spindle

	8.3 Rapid
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Curricular Content Guide

1. Underpinning knowledge	1.1 Identify the major components of a CNC turning center and explain their functions. 1.2 Interpret standard operating procedures for powering the CNC turning center on and off safely. 1.3 Concept of hand jog and Automatic Tool Change (ATC) functions in machine operation. 1.4 Understand correct workpiece setup and tool setup to ensure accuracy and safety. 1.5 Interpret cutting tool geometry and its role in machining accuracy and tool life. 1.6 Understand cutting parameters (speed, feed, depth of cut) based on the job material. 1.7 Identify and use function keys, display keys, mode keys, and override keys for machine adjustments. 1.8 Understand work offset, tool offset, and the application of Visual Programming System (VPS) and Shape Creator.
2. Underpinning skills	2.1 Checking air pressure, coolant, lubrication, and power supply. 2.2 Performing hand jog and Automatic Tool Change (ATC) functions. 2.3 Setting up workpieces and tools for machining. 2.4 Selecting cutting tools and inserts according to machining requirements. 2.5 Choosing appropriate sequence of cutting tools for operations. 2.6 Using function keys, display keys, mode keys, and override keys for machine adjustments. 2.7 Positioning CNC machine axes using hand jog. 2.8 Setting and adjusting work offsets and tool offsets. 2.9 Carrying out visual programming system (VPS) and Shape Creator operations according to drawings.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual)

	4.2 Computer/laptop 4.3 Specifications and work instructions 4.4 Learning materials
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 prepared CNC turning center 1.2 selected cutting tools 1.3 performed CNC turning center using controller
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CREATE PROGRAM USING G & M CODE FOR CNC MACHINE	Nominal Duration: 40 hrs.	Unit Code: SICIP-LE-ACNC-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to create program using G & M code for CNC machine. It specifically includes the tasks of determining machine co-ordinate systems, programing CNC machining center using G & M code, programming CNC turning center using G & M code and testing and debugging CNC program.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Determine Machine Co-Ordinate Systems	1.1 Reference position of the machining center is determined. 1.2 Work coordinates are set to align the workpieces with the required position. 1.3 Absolute & Incremental positioning systems are identified and applied as per job requirement. 1.4 <u>Cutting parameters</u> are calculated according to the job requirements. 1.5 Machine coordinate system is verified to ensure it matches the specifications for the job.
2. Program CNC Machining Center using G & M Code	2.1 <u>G-code</u> and <u>M-code</u> programming format and basic commands are understood and applied. 2.2 Cutter radius compensation and tool length compensation are described. 2.3 <u>Canned cycles for machining center</u> are executed enabling repetitive machining tasks. 2.4 Programs for face milling, pocket milling, and contour milling are written. 2.5 Sub-programs and loops are executed.
3. Program CNC Turning Center using G & M Code	3.1 G-codes and M-codes are understood including their specific functions. 3.2 Structure of a CNC program is explained, including the sequence of commands. 3.3 Tool offsets and TNRC (Tool Nose Radius Compensation) are explained. 3.4 Maximum spindle speed and <u>constant surface speed</u> are explained. 3.5 <u>Canned cycles for turning center</u> are executed, enabling common turning operations. 3.6 VPS (Visual Programming System) and Shape Creator programming are executed.
4. Test and Debug CNC	4.1 CNC program is tested by running it in a safe, dry run

Program	mode.
	4.2 Machine's movements are monitored during the test.
	4.3 Any errors in the CNC program are identified and corrected.
	4.4 Program is debugged by adjusting to optimize performance and ensure safety.
	4.5 Program is re-tested after adjustments to verify that all issues have been resolved.
	4.6 Final adjustments are made to ensure the program meets the required specifications.
	4.7 CNC program is documented with the changes made during the test and debug process.

Range of Variables

Variable	Range
	May include but not limited to:
1. Cutting parameters	1.1 Cutting speed 1.2 Spindle speed 1.3 Feed rate 1.4 Depth of cut
2. G-code	2.1 G00 – Rapid positioning 2.2 G01 – Linear interpolation (controlled feed) 2.3 G02 – Clockwise circular interpolation 2.4 G03 – Counterclockwise circular interpolation 2.5 G04 – Dwell (pause for a specified time) 2.6 G09 – Exact stop (motion control) 2.7 G17 – XY plane selection 2.8 G18 – ZX plane selection 2.9 G19 – YZ plane selection 2.10 G20 – Programming in inches 2.11 G21 – Programming in millimeters 2.12 G28 – Return to machine home position 2.13 G30 – Return to secondary home position 2.14 G31 – Probe for position or measurement 2.15 G32 – Thread cutting 2.16 G33 – Thread cutting with constant pitch 2.17 G40 – Cancel cutter radius compensation 2.18 G41 – Cutter compensation left 2.19 G42 – Cutter compensation right 2.20 G43 – Tool length compensation positive 2.21 G44 – Tool length compensation negative 2.22 G49 – Cancel tool length offset 2.23 G53 – Machine coordinate system 2.24 G54-G59 – Work coordinate systems 2.25 G80 – Canned cycle cancels 2.26 G81- Continuous drill cycle

	2.27 G82- Spot drill cycle 2.28 G83- Pack drill cycle 2.29 G84- Tap cycle 2.30 G90- Absolute positioning/programming system 2.31 G91- Incremental positioning/programming system 2.32 G94- Feed in mm per minute 2.33 G95- Feed in rev per minute 2.34 G98- Retract initial point 2.35 G99- Retract reference point
3. M-code	3.1 M00 – Program stop 3.2 M01 – Optional program stops 3.3 M02 – End of program 3.4 M03 – Spindle on clockwise 3.5 M04 – Spindle on counterclockwise 3.6 M05 – Spindle stop 3.7 M06 – Tool change 3.8 M07 – Coolant on (mist) 3.9 M08 – Coolant on (flood) 3.10 M09 – Coolant off 3.11 M30 – End of program and reset 3.12 M98 - Sub program call 3.13 M99 - Sub program cancel
4. Canned cycles for machining center	4.1 Drilling Cycle (G81) 4.2 Drilling Cycle with Dwell (G82) 4.3 Peck Drilling Cycle (G83) 4.4 Tapping Cycle (G84) 4.5 Boring Cycle (G85)
5. Constant surface speed	5.1 G96 – Constant Surface Speed (CSS) Mode 5.2 G97 – Cancel Constant Surface Speed (CSS) Mode
6. Canned cycles for turning center	6.1 Finish Turning Cycle (G70) 6.2 OD/ID Stock Removal Cycle (G71) 6.3 Facing Cycle (G72) 6.4 Peck Drilling Cycle (G73) 6.5 Grooving Cycle (G75) 6.6 Thread Cutting Cycle (G76) 6.7 Boring Cycle (G85)

Curricular Content Guide

1. Underpinning knowledge	1.1 Concept of reference position, machine coordinate system, and work coordinate system in CNC machining. 1.2 Identify and interpret absolute and incremental positioning systems and their applications. 1.3 Understand the principles of calculating cutting parameters based on job requirements. 1.4 Interpret the structure of CNC programs.
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	1.5 Understand cutter radius compensation, tool length compensation, and tool nose radius compensation (TNRC). 1.6 Understand and apply programming methods. 1.7 Interpret spindle speed concepts. 1.8 Understand the process of testing CNC programs using dry run mode. 1.9 Concept of debugging, re-testing, and documenting CNC programs.
2. Underpinning skills	2.1 Determining reference position of the machining center. 2.2 Setting work coordinates to align workpieces accurately. 2.3 Calculating cutting parameters according to job requirements. 2.4 Verifying machine coordinate system against job specifications. 2.5 Executing canned cycles for milling and turning operations. 2.6 Executing sub-programs and loops. 2.7 Setting and adjusting tool offsets and TNRC (Tool Nose Radius Compensation). 2.8 Executing VPS and Shape Creator programming. 2.9 Testing CNC program in a safe, dry-run mode. 2.10 Monitoring machine movements during testing. 2.11 Debugging programs to optimize performance and ensure safety. 2.12 Re-testing adjusted programs to verify correctness. 2.13 Making final adjustments to meet required specifications.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	The Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Learning manual 4.4 Job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 determined machine co-ordinate systems 1.2 programed CNC machining center using G & M code 1.3 programed CNC turning center using G & M code 1.4 tested and debugged CNC program
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM 2D MACHINING OPERATION FOR CNC MACHINING CENTER	Nominal Duration: 65 hrs.	Unit Code: SICIP-LE-ACNC-04-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to perform 2D machining operation for CNC machining center. It specifically includes the tasks of set-up machine and workpiece, creating/inputting program and simulate the program, operating CNC machining center for 2D machining and performing basic CNC maintenance.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Set-up Machine and Workpiece	1.1 <u>Personal Protective Equipment (PPE)</u> is collected and worn as per job requirement. 1.2 <u>Cutting tools</u> are set on the CNC machine according to the required sequence of operations for 2D machining. 1.3 <u>Blank/workpiece</u> is collected as per job requirement 1.4 <u>Work holding and clamping devices</u> are securely tightened according to standard operating procedures. 1.5 The workpiece is properly mounted and dialed on the clamping devices. 1.6 Work offset and tool offset are set according to the required tool position and workpiece alignment.
4. Create/input Program and Simulate the Program	2.1 Feeds and speeds for milling, drilling and tapping are calculated. 2.2 A program is created on the machine based on the job requirements. 2.3 Program is input into the machine using appropriate devices. 2.4 A program simulation is performed to check the desired tool path movement. 2.5 A dry run is performed as required to verify that the machine.
3. Operate CNC Machining Center for 2D Machining	3.1 Facing operation is performed to create a flat surface on the workpiece. 3.2 Contour operation is performed to shape the workpiece along a defined path or contour. 3.3 <u>Canned cycles</u> are performed to carry out repetitive operations efficiently. 3.4 Pocket milling operations, both rectangular and circular, are performed. 3.5 Workpiece dimensions are checked as per drawing or job requirement.
4. Perform Basic CNC	4.1 Routine maintenance practices are followed.

Maintenance	4.2 Lubrication is periodically checked to ensure all moving parts are properly lubricated. 4.3 Air pressure is checked. 4.4 Alarms are cleared, and troubleshooting is performed. 4.5 Tools and equipment are cleaned and stored as per standard operating procedures.
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Range of Variables

Variable	Range May include but not limited to:
1. Personal Protective Equipment (PPE)	1.1 Apron 1.2 Mask 1.3 Safety goggles 1.4 Ear plug 1.5 Safety helmet 1.6 Safety shoes
2. Cutting tools	2.1 Face mill 2.2 End mill 2.3 Ball mill 2.4 Bull mill 2.5 Chamfer mill 2.6 Machine tap 2.7 Drill bit 2.8 Centre drill
3. Blank/workpiece	3.1 Mild steel block 3.2 Aluminum block
4. Work holding and clamping devices	4.1 Vices 4.2 Clamping kits 4.3 Chucks 4.4 T-Nuts & T-Slots 4.5 Angle plate 4.6 V-Block 4.7 Parallel bar
5. Canned cycles	5.1 Drilling Cycle 5.2 Drilling Cycle with Dwell 5.3 Peck Drilling Cycle 5.4 Tapping Cycle

Curricular Content Guide

1. Underpinning knowledge	1.1 Understand the sequence of setting cutting tools for 2D machining operations. 1.2 Identify proper work holding and clamping devices, and interpret standard procedures for secure setup. 1.3 Interpret the process of mounting, aligning, and dialing a workpiece on clamping devices.
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	1.4 Understand work offset and tool offset principles. 1.5 Concept of feeds, speeds, and their calculation for efficient machining performance. 1.6 Understand CNC program creation, input methods, simulation, and dry run testing to verify tool paths. 1.7 Interpret common machining operations and their applications. 1.8 Understand dimensional inspection methods to verify workpiece accuracy against job requirements.
2. Underpinning skills	2.1 Collecting blanks/workpieces as per job requirements. 2.2 Securing work holding and clamping devices according to procedures. 2.3 Mounting and dialing the workpiece on clamping devices. 2.4 Calculating feeds and speeds for milling, drilling, and tapping. 2.5 Creating and inputting programs into the CNC machine. 2.6 Performing program simulation and dry runs to verify tool paths. 2.7 Performing facing, contour, and pocket milling operations. 2.8 Executing canned cycles for repetitive machining tasks. 2.9 Checking workpiece dimensions against drawings or specifications. 2.10 Following routine maintenance and lubrication checks. 2.11 Checking air pressure and clearing machine alarms.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Learning manual 4.4 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate:
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	1.1 set-up machine and workpiece 1.2 created/inputted program and simulate the program 1.3 operated CNC machining center for 2D machining 1.4 performed basic CNC maintenance
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM 2D MACHINING OPERATION FOR CNC TURNING CENTER	Nominal Duration: 65 hrs.	Unit Code: SICIP-LE-ACNC-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to perform 2D machining operation using CNC turning center. It specifically includes the tasks of set-up machine cutting tools and workpiece, creating/inputting program and simulate the program, operate CNC turning center for 2D machining.		

Elements and Performance Criteria

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Set-up Machine Cutting tools and Workpiece	1.1 <u>Personal Protective Equipment (PPE)</u> is collected and worn as per job requirement. 1.2 <u>Cutting tools</u> and <u>inserts</u> are set on the CNC machine according to the required sequence of operations for 2D machining. 1.3 <u>Blank/workpiece</u> is collected as per job requirement 1.4 Jaws of the chuck are adjusted as per workpiece diameter. 1.5 The workpiece is properly mounted on the chuck. 1.6 Work offset is set according to the required tool position and workpiece alignment.
2. Create/input Program and Simulate the Program	2.1 Feeds and speeds for turning, drilling, boring and threading are calculated. 2.2 A program is created on the CNC turning center based on the job requirements. 2.3 Program is input into the CNC turning center using the <u>appropriate devices</u> . 2.4 Program is simulated to verify the tool path movement. 2.5 Any issues found during the simulation are corrected. 2.6 Simulation is re-checked after adjustments.
3. Operate CNC Turning Center for 2D Machining	3.1 Facing operation is performed to create end face on the workpiece. 3.2 OD/ID stock removal cycle is performed as per job requirement. 3.3 Drilling and boring operation is carried out as per dimension. 3.4 Internal and external threading operation is carried out as per drawing. 3.5 Grooving/parting operation is carried out as per job dimension. 3.6 Workpiece measurement is checked as per job dimension.

Range of Variables

Variable	Range May include but not limited to:
1. Personal Protective Equipment (PPE)	1.1 Apron 1.2 Mask 1.3 Safety goggles 1.4 Ear plug 1.5 Safety helmet 1.6 Safety shoes
2. Cutting tools	2.1 Facing tool 2.2 Turning tool 2.3 Grooving/ parting tool 2.4 Threading tool 2.5 Boring tool 2.6 Drill bit 2.7 Center drill
3. Inserts	3.1 CNMG 3.2 DNMG 3.3 TNMG 3.4 VNMG 3.5 WNMG 3.6 Threading insert 3.7 Parting insert
4. Blank/workpiece	4.1 Mild steel shaft 4.2 Aluminum shaft 4.3 Teflon shaft
5. Appropriate devices	5.1 Computer/laptop 5.2 USB drive 5.3 Flash drive 5.4 Card readers 5.5 LAN connection

Curricular Content Guide

1. Underpinning knowledge	1.1 Identify cutting tools and inserts 1.2 Understand the correct sequence of setting them for 2D turning operations. 1.3 Interpret chuck jaw adjustment and proper mounting methods to secure the workpiece accurately. 1.4 Understand work offset principles and their role in aligning tools and workpieces for precision machining. 1.5 Understand CNC program creation, input methods, simulation, and verification of tool path. 1.6 Interpret common turning operations. 1.7 Understand dimensional inspection and measurement techniques to ensure workpiece accuracy.
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	1.8 Concept of correcting simulation issues and rechecking programs to ensure safe and precise machining.
2. Underpinning skills	2.1 Collecting blanks/workpieces as per job requirements. 2.2 Adjusting jaws of the chuck according to workpiece diameter. 2.3 Mounting the workpiece properly on the chuck. 2.4 Setting work offsets for tool position and workpiece alignment. 2.5 Calculating feeds and speeds for turning, drilling, boring, and threading. 2.6 Creating and inputting programs into the CNC turning center. 2.7 Simulating programs and correcting issues found during simulation. 2.8 Performing facing, OD/ID stock removal, drilling, and boring operations. 2.9 Executing internal and external threading, grooving, and parting operations. 2.10 Checking workpiece measurements against job specifications.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 set-up machine cutting tools and workpiece 1.2 created/inputted program and simulate the program 1.3 operated CNC turning center for 2D machining
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration

	2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INTERPRET CO-ORDINATE MEASURING MACHINE (CMM)	Nominal Duration: 20 hrs.	Unit Code: SICIP-LE-ACNC-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to interpret co-ordinate measuring machine (CMM). It specifically includes the tasks of understanding the concept of CMM, understanding CMM program and operation and analyzing data.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand the Concept of CMM	1.1 Concept of a Coordinate Measuring Machine (CMM) is understood. 1.2 <u>Components of the CMM</u> are recognized for their role in accurate measurement. 1.3 <u>Types of probes</u> are identified and understood based on their applications in different measurement scenarios. 1.4 Importance of calibration in CMM operation is understood.
2. Understand CMM Program and Operation	2.1 CMM software interface is understood. 2.2 Function of CMM software is explained to generate precise measurements. 2.3 <u>Feature measurement basis</u> of the CMM is explained. 2.4 <u>CMM settings</u> is understood for a specific measurement task. 2.5 Measurement cycle is explained to measure <u>Geometric Dimension and Tolerance (GD&T)</u> .
3. Analyze Data	3.1 Measurement results are compared against the part's specifications. 3.2 Measurement data is interpreted and any out-of-tolerance results are identified. 3.3 Trends and patterns in the data are examined and highlighted. 3.4 Inspection report is generated and reported.

Range of Variables

Variable	Range
	May include but not limited to:
1. Components of the CMM	1.1 Framework 1.2 Measuring axes 1.3 Probes system

	1.4 Granite table 1.5 Controller/Drive system 1.6 Calibration and feedback system 1.7 Calibration master
2. Types of probes	2.1 Touch Trigger Probe (TTP) 2.2 Scanning 2.3 Optical/laser
3. Feature measurement basis	3.1 Point 3.2 Line 3.3 Circle 3.4 Plane 3.5 Cylinder 3.6 Sphere
4. CMM settings	4.1 Probe calibration 4.2 Part alignment 4.3 Datum setup
5. Geometric Dimension and Tolerance (GD&T)	5.1 Straightness 5.2 Flatness 5.3 Circularity 5.4 Cylindricity 5.5 Profile of line 5.6 Profile of surface 5.7 Angularity 5.8 Perpendicularity 5.9 Parallelism 5.10 Circular run-out 5.11 Total run-out 5.12 Position concentricity 5.13 Symmetry

Curricular Content Guide

1. Underpinning knowledge	1.1 Concept of a Coordinate Measuring Machine (CMM) and its role in precision measurement. 1.2 Identify the major components of a CMM and understand their functions in accurate measurement. 1.3 Understand the types of probes and their applications for different measurement tasks. 1.4 Interpret the importance of calibration to ensure reliable and precise measurements. 1.5 Understand the CMM software interface and its functionalities for measurement operations. 1.6 Explain how CMM software generates precise measurement data for parts. 1.7 Interpret the feature-based measurement approach for Geometric Dimensioning and Tolerancing (GD&T). 1.8 Understand CMM settings and configurations for specific
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	measurement tasks. 1.9 Analyze measurement results, identify out-of-tolerance conditions, and examine trends or patterns in data. 1.10 Concept of generating, documenting, and reporting inspection results for quality assurance purposes.
2. Underpinning skills	2.1 Performing calibration to ensure accurate CMM operation. 2.2 Using the CMM software interface for measurement tasks. 2.3 Operating CMM software to generate precise measurements. 2.4 Measuring features according to the part's specifications. 2.5 Setting up CMM for specific measurement tasks. 2.6 Executing measurement cycles for GD&T inspection. 2.7 Comparing measurement results against specifications. 2.8 Examining trends and patterns in measurement data. 2.9 Generating and reporting inspection results.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 understood the concept of CMM 1.2 understood CMM program and operation 1.3 analyzed data
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)

3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Advanced CNC Machining Center Operation and Practice
Number of Trainees:	25

Course-wise Training Space:

- Computer Lab : 400 sft
- Workshop : 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Computer/Laptops	01
2.	Printer	01
3.	Multimedia projector	01
4.	Internet connectivity with power source	01
5.	CNC Machining Centers	05
6.	Tool Grinding Machine	01
7.	Hand Grinder	01
8.	Power Saw	01
9.	Hand Drill Machine	01
10.	CNC Turning Centers	05
11.	Co-ordinate Measuring Machine (CMM)	01
12.	Tool Holders and Tool Kits	05
13.	CNC Machine Setup and Calibration Tools	05
14.	Cutting Tools Set	05
15.	Work holding Devices	05
16.	Measuring Instruments	05
17.	Tool Pre-setter	05

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Advanced CNC Machining Center Operation and Practice, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.